

Monoalphabetic Substitution Cipher

2. Write a C program for monoalphabetic substitution cipher maps a plaintext alphabet to a ciphertext alphabet, so that each letter of the plaintext alphabet maps to a single unique letter of the ciphertext alphabet.

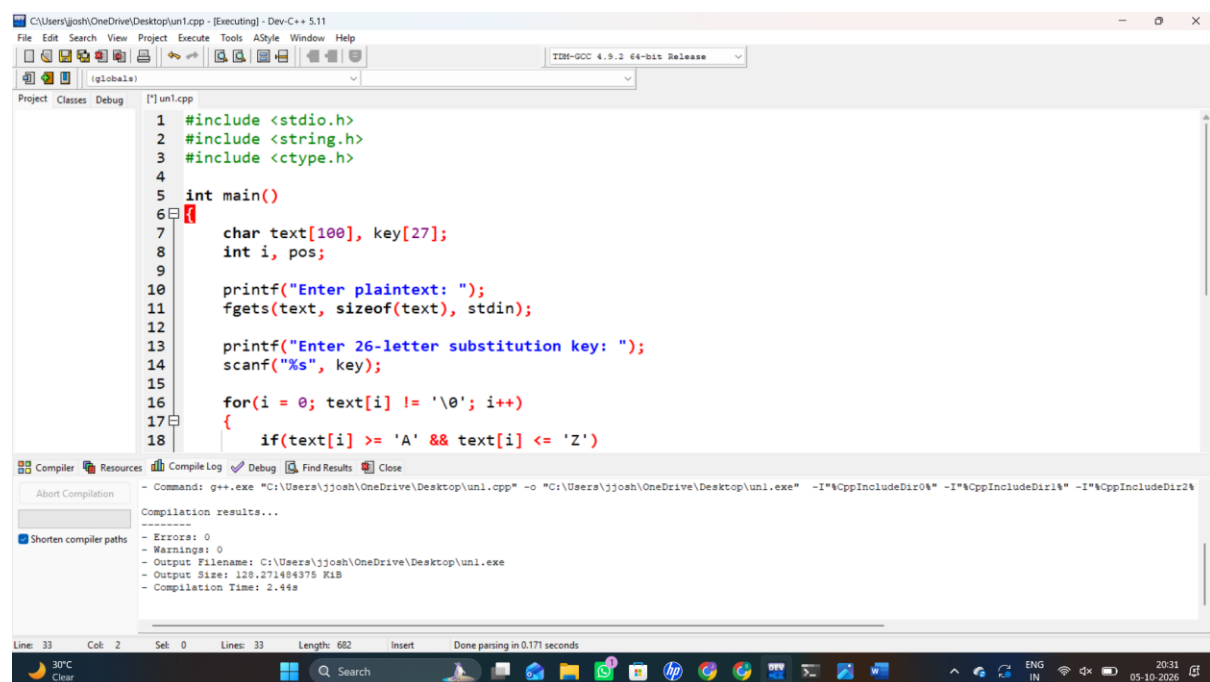
Aim

To write a C program to implement a **Monoalphabetic Substitution Cipher**, where each plaintext letter is mapped to one unique ciphertext letter.

Algorithm

1. Start the program.
2. Read the plaintext.
3. Read the 26-letter substitution key.
4. For every character in the plaintext:
 - Find its position in the alphabet.
 - Replace it with the corresponding character from the substitution key.
5. Preserve spaces and other characters.
6. Display the ciphertext.
7. Stop.

Program



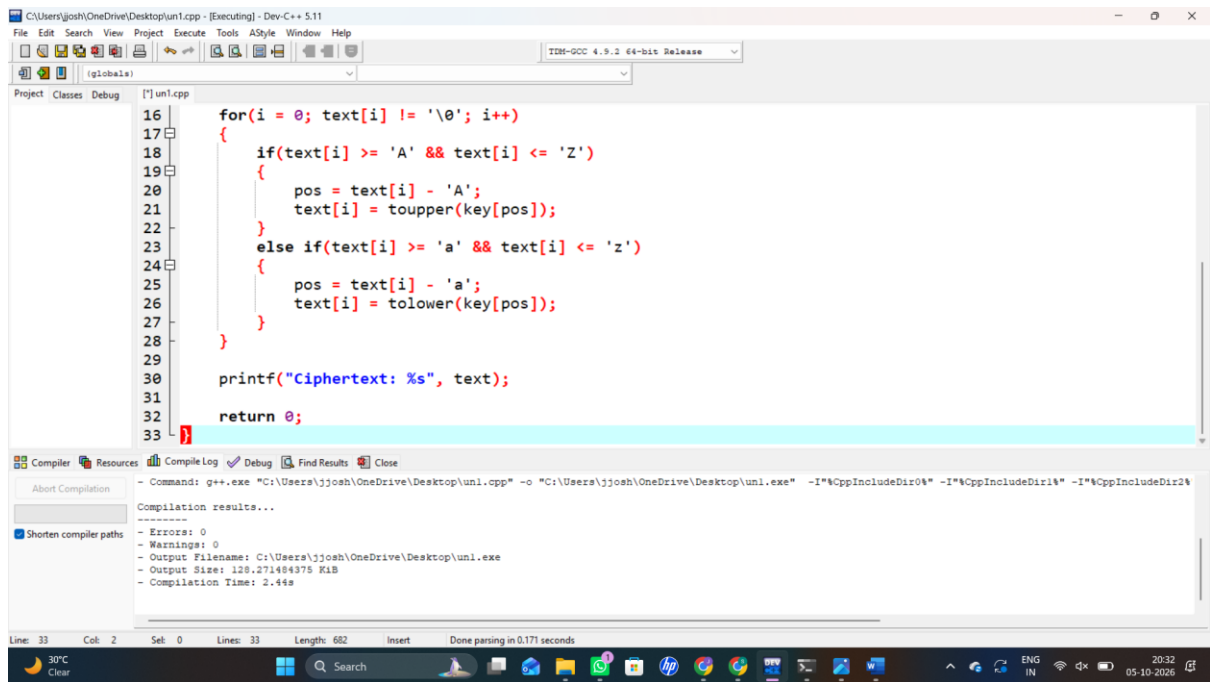
The screenshot shows a Dev-C++ IDE window titled "C:\Users\josh\OneDrive\Desktop\un1.cpp - [Executing] - Dev-C++ 5.11". The code in the editor is as follows:

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 int main()
6 {
7     char text[100], key[27];
8     int i, pos;
9
10    printf("Enter plaintext: ");
11    fgets(text, sizeof(text), stdin);
12
13    printf("Enter 26-letter substitution key: ");
14    scanf("%s", key);
15
16    for(i = 0; text[i] != '\0'; i++)
17    {
18        if(text[i] >= 'A' && text[i] <= 'Z')
```

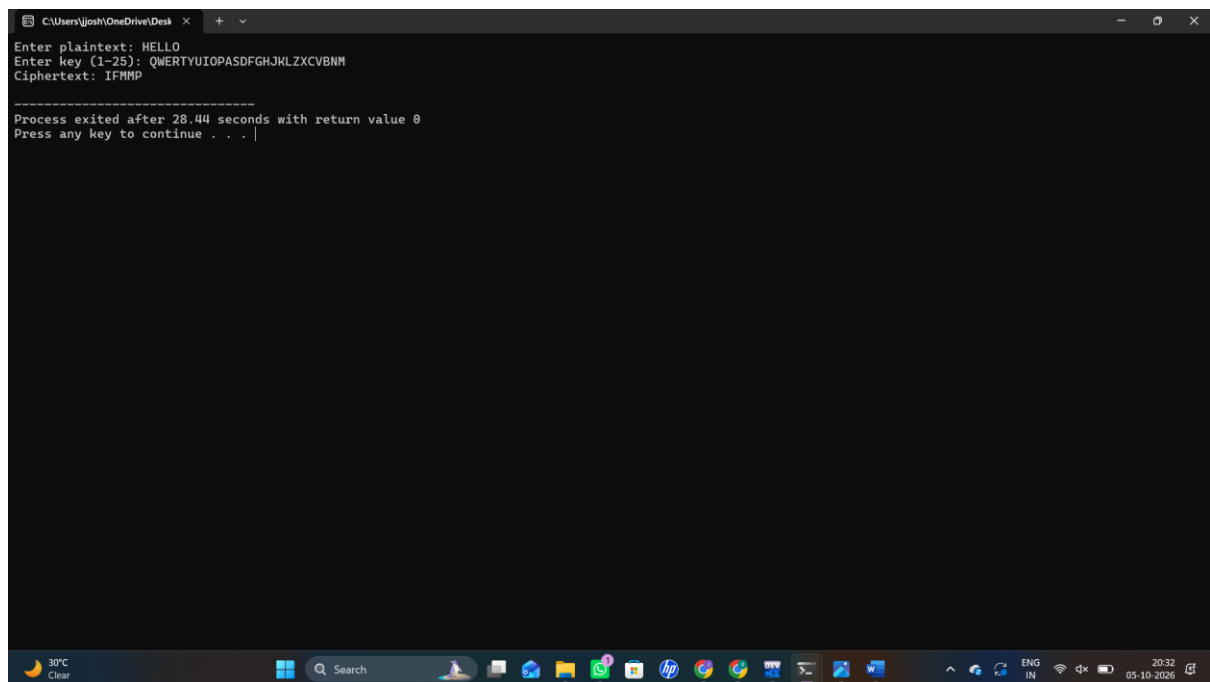
The bottom panel shows the "Compiler" tab with the following output:

```
- Command: g++.exe "C:\Users\josh\OneDrive\Desktop\un1.cpp" -o "C:\Users\josh\OneDrive\Desktop\un1.exe" -I"%CpIncludeDir%" -I"%CpIncludeDir2%" -I"%CpIncludeDir3%"
- Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\josh\OneDrive\Desktop\un1.exe
- Output Size: 128.27184637% R1B
- Compilation Time: 2.44s
```

The status bar at the bottom indicates "Line: 33, Col: 2, Sel: 0, Lines: 33, Length: 682, Insert, Done parsing in 0.171 seconds". The system tray shows a temperature of 30°C, a search bar, and the date 05-10-2026.



Input/Output



Result

Thus, the Monoalphabetic Substitution Cipher was successfully implemented by mapping each plaintext alphabet to a unique ciphertext alphabet.